Direct Matérn Portfolio Policy

Methodology note for the Jensen-Kelly-Pedersen Common Task Framework (CTF)
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1. Objective

The model learns portfolio weights directly from stock characteristics rather than first estimating stock-level expected returns and then solving a separate portfolio problem. At month *t*, the input is the contemporaneously observed characteristic vector $x_{i,t}$. The model returns a cross-sectional portfolio weight for each stock. The estimation objective is the response-one criterion, which is a maximum-Sharpe regression representation for a linear portfolio class.

$$min_f\ (1/T)\ sum_t\ [1 - R^p_{t+1}(f)]^2 + lambda\ ||f||^2$$

The submission uses a Matérn-3/2 characteristic kernel represented by a fixed random-feature map. The random draw is seeded once and is therefore deterministic.

2. Characteristic preprocessing

All transformations are performed separately within the contemporaneous cross section. For each characteristic, the model computes a cross-sectional percentile rank and centers it around zero. Missing values are mapped to zero after centering. Exact raw zeros are retained as a special value. No full-sample mean, variance, feature ranking, or test-period performance enters the preprocessing.

3. Matérn feature map and managed returns

Let phi(x) denote the fixed P-dimensional random-feature approximation to the Matérn-3/2 kernel. For month t, after the next-month excess returns are realized, the historical managed-feature return is

$$F_t = (1/N_t)\ sum_i\ r^e_{i,t+1}\ phi(x_{i,t}).$$

For a coefficient vector beta, the portfolio return equals F_t' beta. Thus the entire cross section enters the estimation only through historical managed returns. The stock-level weight at decision time t is

$$w_{i,t} = phi(x_{i,t})'\ beta\text{-}hat / N_t.$$

4. Response-one ridge estimator

Using the most recent T=120 completed monthly managed returns, the estimator solves

$$beta\text{-}hat = (F'F/T + lambda\ I)^{-1}\ F'1/T.$$

The CTF submission fixes P=512, Matérn smoothness nu=3/2, and a conservative ridge multiplier before any test-period evaluation. The kernel length scale is estimated from observations strictly preceding the first CTF test month using a median-distance rule. No leaderboard result is used for model selection or tuning.

5. Temporal integrity

The model is evaluated sequentially. At a decision month t, beta-hat is computed only from managed returns whose payoffs have already been realized. The current value $r^e_{i,t+1}$ is not added to the history until after the time-t portfolio weights have been formed. Consequently, changing current or future return observations does not change the portfolio weights already produced at time t.

> **The implementation is tested by truncation: the time-t weights from a full dataset must match the time-t weights obtained when every later observation is removed. A second test shocks all current and future lead returns and verifies that current weights are unchanged.**

6. Effective portfolio complexity used in the paper

The paper's empirical complexity analysis uses the same Matérn family but varies ridge regularization. Complexity is measured from the economic managed-payoff operator, not from the raw characteristic kernel. With $G_T = F F'/T$, the empirical effective dimension is

$$\hat{C}_T(\lambda) = \mathrm{tr}\{ G_T [G_T + \lambda I]^{-1} \}.$$

This distinction is central: nominal feature dimension P can be very large while only a much smaller number of portfolio directions are statistically active in a finite market history. The paper therefore reports out-of-sample Sharpe as a function of this managed-payoff effective dimension, the growth of the Sharpe-maximizing effective dimension with sample size, the in-sample/out-of-sample complexity wedge, and nominal versus effective complexity.

7. CTF implementation summary

Component	Submission choice
Language / runtime	Python 3.13
Inputs	CTF chars, feature list, daily returns argument
Feature class	Matérn-3/2 random features
Nominal dimension	P = 512
Training history	Trailing 120 completed months
Regularization	Fixed ridge multiplier; no test-period tuning
Output	id, eom, w
Randomness	Fixed deterministic seed
External data / network	None

8. References

Britten-Jones, M. (1999). The Sampling Error in Estimates of Mean-Variance Efficient Portfolio Weights. Journal of Finance 54(2), 655-671.

Jensen, T. I., Kelly, B., and Pedersen, L. H. (2023). Is There a Replication Crisis in Finance? Journal of Finance 78(5), 2465-2518.

Jensen-Kelly-Pedersen Common Task Framework, rules and data specification, accessed September 4, 2026.

Performance statement. This document describes the submitted algorithm only. It contains no CTF test-period performance claim. Performance should be reported only after the official CTF evaluation has completed.